

Hearing Aids - Medical Clinical Policy Bulletins

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Policy

Scope of Policy

This Clinical Policy Bulletin addresses air conduction and implantable hearing aids.

1. Medical Necessity

Aetna considers the following medically necessary:

1. Air conduction hearing aids when the following criteria are met:
 1. Hearing thresholds 40 decibels (dB) HL or greater at 500, 1000, 2000, 3000, or 4000 hertz (Hz); *or*
 2. Hearing thresholds 26 dB HL or greater at three of these frequencies; *or*
 3. Speech recognition less than 94 percent;
2. Implantable hearing aids (e.g., the Esteem fully implanted active middle ear implant and the Carina prosthesis) and semi-implantable hearing aids (e.g., the Maxum system and the Vibrant Soundbridge) for members who have moderate-to-severe sensorineural hearing impairment and can not tolerate an ear mold because of medical conditions (such as auricular atresia or severe chronic otitis externa);
3. The Bose Hearing Aid and other FDA-cleared hearing aids available over the counter without a prescription as equally effective alternatives to hearing aids available only by prescription for persons whose hearing has been evaluated and meet medical necessity criteria for air conduction hearing aids, and the member has a prescription for the hearing aid from a physician or provider licensed to prescribe hearing aids.

2. Experimental, Investigational, or Unproven

Aetna considers the following procedures experimental, investigational, or unproven because the effectiveness of these approaches has not been established:

1. Air conduction hearing aids for improvement of balance
2. Cartilage conduction hearing aids
3. Hearing aids and semi-implantable hearing aids for all other indications (including for improvement of depression and cognitive decline in the elderly)

4. Use of free-floating piezoelectric microphone in an implantable hearing aid.

3. Policy Limitations and Exclusions

Most Aetna benefit plans exclude coverage of hearing aids. Any applicable benefit plan exclusions and limitations for coverage of hearing aids would apply to air conduction hearing aids, implantable hearing aids and semi-implantable hearing aids. Please check benefit plan descriptions for details.

For plans that do not exclude hearing aids, either OTC and prescription hearing aids are eligible for coverage if they are cleared by the FDA and prescribed by a qualified healthcare provider and medical necessity criteria for hearing aids above are met.

4. Related Policies

- [CPB 0403 - Bone-Anchored Hearing Aids](#)
 - [CPB 0406 - Tinnitus Treatments](#)
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CPT Codes / HCPCS Codes / ICD-10 Codes

Code	Code Description
CPT codes covered when selection criteria are met:	
0951T	Totally implantable active middle ear hearing implant; initial placement, including mastoidectomy, placement of and attachment to sound processor
0952T	Totally implantable active middle ear hearing implant; revision or replacement, with mastoidectomy and replacement of sound processor
0953T	Totally implantable active middle ear hearing implant; revision or replacement, without mastoidectomy and replacement of sound processor
0954T	Totally implantable active middle ear hearing implant; replacement of sound processor only, with attachment to existing transducers
0955T	Totally implantable active middle ear hearing implant; removal, including removal of sound processor and all implant components
Other CPT codes related to the CPB:	
92628	Evaluation for hearing aid candidacy, unilateral or bilateral, including review and integration of audiologic function tests, assessment, and interpretation of hearing needs (eg, speech-in-noise, suprathreshold hearing measures), discussion of candidacy results, counseling on treatment options with report, and, when performed, assessment of cognitive and communication status; first 30 minutes
92629	each additional 15 minutes (List separately in addition to code for primary procedure)
92631	Hearing aid selection services, unilateral or bilateral, including review of audiologic function tests and hearing aid candidacy evaluation, assessment of visual and dexterity limitations, and psychosocial factors, establishment of device type, output requirements, signal processing strategies and additional features, discussion of device recommendations with report; first 30 minutes
92632	each additional 15 minutes (List separately in addition to code for primary procedure)

Code	Code Description
92634	Hearing aid fitting services, unilateral or bilateral, including device analysis, programming, verification, counseling, orientation, and training, and, when performed, hearing assistive device, supplemental technology fitting services; first 60 minutes
92635	each additional 15 minutes (List separately in addition to code for primary procedure)
92636	Hearing aid post-fitting follow-up services, unilateral or bilateral, including confirmation of physical fit, validation of patient benefit and performance, sound quality of device, adjustment(s) (eg, verification, programming adjustment[s], device connection[s], and device training), as indicated, and, when performed, hearing assistive device, supplemental technology fitting services; first 30 minutes
92637	each additional 15 minutes (List separately in addition to code for primary procedure)
92638	Behavioral verification of amplification including aided thresholds, functional gain, speech-in-noise, when performed (List separately in addition to code for primary procedure)
92639	Hearing-aid measurement, verification with probe-microphone (List separately in addition to code for primary procedure)
92641	Hearing device verification, electroacoustic analysis
92642	Hearing assistive device, supplemental technology fitting services (eg, personal frequency modulation [FM]/digital modulation [DM] system, remote microphone, alerting devices)
HCPCS codes covered if selection criteria are met::	
S2230	Implantation of magnetic component of semi-implantable hearing device on ossicles in middle ear
V5030	Hearing aid, monaural, body worn, air conduction
V5050	Hearing aid, monaural, in the ear
V5070	Glasses, air conduction
V5095	Semi-implantable middle ear hearing prosthesis
HCPCS codes not covered for indications listed in the CPB:	
<i>Cartilage conduction hearing aids - no specific code</i>	
ICD-10 codes covered if selection criteria are met::	
H90.3 - H90.8	Sensorineural hearing loss and mixed conductive and sensorineural hearing loss
Q16.1	Congenital absence, atresia and stricture of auditory canal (external)
ICD-10 codes not covered for indications listed in the CPB:	

Code	Code Description
F32.0 - F32.9	Major depressive disorder, single episode
F33.0 - F33.9	Major depressive disorder, recurrent
H81.311 - H81.399	Other peripheral vertigo
H81.8X1 - H81.8X9	Other disorders of vestibular function
H81.90 - H81.93	Unspecified disorder of vestibular function
H82.1 - H82.9	Vertiginous syndromes in diseases classified elsewhere
R41.81	Age-related cognitive decline

Background

Implantable hearing systems (middle ear implants), in which all components are inserted within an individual, are purported to be an alternative to the traditional over the ear hearing aid. Examples of totally implantable hearing systems include, but may not be limited to, Esteem totally implantable hearing system and Carina fully implantable hearing device.

Partially implantable or semi implantable electromagnetic hearing aids, also called bone conduction hearing aids, create an electromagnetic field that vibrates and stimulates the ossicles, sending signals to the cochlea. Examples of semi-implantable electromagnetic hearing aids include, but may not be limited to, Vibrant Soundbridge system, Maxum system, Otomag alpha 1(M) bone conduction hearing system and Semi-Implantable middle ear transducer (MET) ossicular stimulator system.

Vibrant Soundbridge

According to the manufacturer, the Vibrant Soundbridge is not a cochlear implant, but an implantable alternative to an external hearing aid for adults who have moderate-to-severe sensorineural hearing loss. Unlike the cochlear implant, the Vibrant Soundbridge is not a prosthetic replacement for the ear; rather, the Vibrant Soundbridge acts as a hearing aid in amplifying sounds, and like a hearing aid, is indicated for persons with sensorineural hearing loss.

The Vibrant Soundbridge consists of 2 components:

The AP picks up sound from the environment and transmits that sound across the skin to the implanted receiver. The implanted receiver converts the signal and transmits it to the FMT, which is a transducer that directly vibrates the ossicles by mimicking the natural motion of the ossicular chain, sending an enhanced signal to the fluid-filled inner ear (cochlea). The ossicular motion creates movement in the cochlea stimulating the hair cells, which in turn provide stimuli to the auditory nerve, which are then interpreted by the brain as sound.

Snick et al (2001) explains that "application of the Vibrant Soundbridge involves surgery and is more expensive than the application of conventional hearing aids. Therefore, to justify its use, the results obtained with the Vibrant Soundbridge should be better than those obtained with conventional hearing aids (except in patients with severe chronic otitis externa who can not tolerate an ear mold)."

The manufacturer states that the Vibrant Soundbridge provides superior results to hearing aids. But the clinical data are inconsistent (e.g., Snik and Cremers, 2001) and the Food

and Drug Administration (FDA) did not allow the manufacturer (Symphonix) to claim superiority to standard hearing aids in product labeling indications. Rather, the product is indicated for patients who "desire an alternative to an acoustic hearing aid." In clinical studies presented to the FDA, measures of patient's subjective preference for the implant were not matched by clinically significant in objective measures between the implant and conventional hearing aids. FDA's Commissioner concluded that "[t]he results showed that the participants could hear about as well with the implant as with traditional hearing aids."

The Vibrant Soundbridge does offer clinically significant benefit to adult patients who have moderate to severe sensorineural hearing loss and severe chronic otitis externa such that they are unable to be fitted with an ear mold. The Vibrant Soundbridge has not been shown to offer clinically significant benefits to other patients with sensorineural hearing loss.

In general, an implantable hearing aid consists of a transducer that is coupled to the ossicular chain and electronics. The coupling is of major importance. The Vibrant Soundbridge transducer is fixed to the ossicular chain by means of a special clip that is crimped around the long process of the incus. In addition to crimping, bone cement has been used to optimize the fixation. Snik and Cremers (2004) reported that there was no negative effect of using bone cement; however, there is also no reason to use bone cement in users of the Vibrant Soundbridge on a regular basis.

Implantable hearing aids have been used in persons with auricular atresia. Kiefer and associates (2006) noted that congenital malformations of the auricle are often combined with atresia of the outer ear canal and malformations of the ossicles, representing aesthetic as well as functional deficits. Optimal treatment should therefore address both aspects equally. These investigators described a new approach that combine the reconstruction of the auricle with implantation of an active middle ear hearing aid, stimulating the round window membrane. A 33-year old man, with bilateral ear microtia, fibrous atresia of the external ear canals and malformation of the ossicles due to Treacher Collins-Franceschetti syndrome was included in the study. In stage I, the cartilage framework of the new auricle, made of autogenous rib cartilage, was fabricated and implanted. During stage II, the auricle was elevated, a retro-auricular sulcus was formed and a Vibrant MED-EL Soundbridge device was implanted. The transducer was coupled to the round window membrane. Both functional as well as aesthetical results were favorable. Aided thresholds were between 15 and 30 decibel (dB) in the frequency range of 0.75 to 6 kHz, mono-syllabic word understanding at 65 dB SPL increased from 0 to 80 %. The authors concluded that combining aesthetic and functional rehabilitation, autogenous reconstruction of a new auricle together with the implantation of an active middle ear hearing aid, coupled to the round window membrane, is a promising new approach.

Siebert and colleagues (2007) noted that patients with congenital auricular atresia suffer from a conductive hearing loss (HL) with an air-bone gap of 50 to 60 dB. Conventional bone conducting or bone anchored hearing aids are treatment options with several disadvantages and a biophysical limitation of almost no sound attenuation in the skull bone. In a prospective study, these investigators attempted to improve the hearing results of auricular atresia with the use of fully implantable hearing aids (Otologics Fully Implantable Middle Ear Transducer). They were implanted in 5 patients with congenital auricular atresia and their audiological outcome evaluated. After activation and fitting of the devices, patients experienced an improvement of sound-field thresholds up to 50 dB HL. The mean functional gain in a three frequency pure-tone average was approximately 35 dB HL. The authors concluded that this technique appears to provide a completely new dimension for the audiological rehabilitation of patients with severe malformation of the middle ear.

Wollenberg et al (2007) presented 3 cases of unilateral atresia to illustrate a combined approach integrating hearing rehabilitation using the active middle ear implant Vibrant Soundbridge (VSB) into plastic auricular reconstruction. The VSB was attached to the stapes supra-structure via the titanium clip in 2 of these cases and in the 3rd case a sub-facial approach was used to attach it directly to the membrane of the round window. The air-bone gap was reduced to 17 dB, 14 dB and 0.25 dB HL. In free-field speech recognition tests at 65 dB SPL, the patients achieved 100 %, 90 % and 100 % recognition with the activated implant. No post-operative complications such as facial nerve paresis, vertigo or inner ear damage were found. The authors concluded that the integration of active middle ear implants in auricular reconstruction opens up a new approach in complete hearing rehabilitation. The additional implantation of the VSB does

not have any negative effect on the healing process or the cosmetic outcome of the auricular reconstruction.

An assessment of the Vibrant Soundbridge and other middle ear implants (MEI) by the Australian Medical Services Advisory Committee (MSAC, 2010) found that "there was a paucity of high level evidence with which to assess the effectiveness of the MEI."

Luers and Huttenbrink (2014) stated that the Vibrant Soundbridge is the world's most often implanted active middle ear implant or hearing aid. During the last few years, the device indications have expanded from sensorineural hearing loss to conductive and mixed hearing loss. Titanium couplers have led to improved contact of the floating mass transducer with the middle ear structures. The resulting hearing gain is satisfying for most patients, but so far, there is no clear audiologic advantage over conventional hearing aids. The authors concluded that the indications are mainly related to intolerance of conventional hearing aids (e.g., chronic otitis externa), severe mixed hearing loss with a destructed middle ear and certain medical diagnosis (e.g., congenital atresia).

Shohet et al (2011) evaluated the effectiveness of the Envoy Esteem totally implantable hearing device in treating profound high-frequency sensorineural hearing loss (PHFSNHL). A total of 5 patients with PHFSNHL participated in a prospective, multi-center, non-randomized Food and Drug Administration clinical trial. Main outcome measures included speech reception threshold and word recognition scores (WRS) at 50 dB HL presentation level. Pre-operative speech reception threshold improved from an unaided 65 dB and aided 48 dB average to 26 dB with the Esteem at 12 months; WRS at 50 dB scores improved from an unaided 10 % and aided 23 % average to 78 % post-operatively. The authors concluded that the Esteem totally implantable middle ear hearing device provided appreciable functional gain and improvement in WRS to rehabilitate hearing in patients with PHFSNHL.

The FDA classifies the Vibrant Soundbridge as a surgically implanted hearing device intended to help adults with moderate-to-severe nerve hearing loss. It is used as an alternative to traditional hearing aids, adults with a moderate-to-severe sensorineural hearing loss may choose this device. Adults who choose this device should have already tried using appropriately fit hearing aids.

Furthermore, the American Speech-Language-hearing Association (ASHA, 2016) lists HCPCS code V5095 (Semi-implantable middle ear hearing prosthesis (use for Vibrant Soundbridge)) for reimbursable audiology related devices.

Totally Implantable Hearing Device

The Esteem Totally Implantable Hearing System by Envoy Medical Corporation (St. Paul, MN) is a fully implantable system that uses piezoelectric transducers. It uses the eardrum as the microphone. The mechanical signal is detected by a piezoelectric transducer at the head of the incus (the sensor) and converted to an electrical signal that is amplified, filtered, and converted back to a vibratory signal. The processed vibratory signal is then delivered by another piezoelectric transducer (the driver) to the stapes capitulum (Altuna Mariezcurrena et al, 2008). The Esteem hearing implant is different from all other microphone-based hearing devices (e.g., hearing aids, other middle ear implants or cochlear implants) because it uses the eardrum to process the incoming sound and thus preserves a natural way of hearing. The Esteem is implanted under the skin behind the ear and in the middle ear space, and therefore invisible.

In a prospective, single-subject, repeated-measures, multi-center, phase I clinical study, Chen and colleagues (2004) evaluated safety and functionality of the Envoy System, a totally implantable middle ear hearing system for sensorineural loss. Data collected included Abbreviated Profile of Hearing Aid Benefit, bone conduction threshold, speech reception threshold, functional gain, word recognition, and adverse events. Testing was performed unaided, with the patient's best-fit hearing aid, and post-device activation at 2 (trial endpoint) and 4 months. Five of 7 patients at the 2-month post-activation period had working systems. All 5 patients perceived benefit increases with the Envoy System over their best-fit hearing aid, including communication in high background noise levels. Word recognition was improved over hearing aids. Functional gain and speech reception thresholds were similar for the Envoy device and hearing aids. The authors concluded that the feasibility trial has shown the Envoy device can safely sense and drive the ossicular chain.

Barbara et al (2009) evaluated the benefits of a totally implantable middle ear device,

the Esteem 2 (Envoy Medical), in subjects affected by moderate-to-severe sensorineural hearing loss as measured through pure tone audiometry testing carried out during the different post-operative fitting sessions. A total of 6 patients were included in this study. Selection was carried out via pre-operative audiometric tests and thorough counseling, which considered information on previous experience with conventional hearing aids as well as each patient's motivation to undergo a surgical application. Specific surgical training is needed to accommodate routine surgical steps along with less familiar steps, such as placement of cement material and overall fixation of the system. The surgical procedure took a long time, but a reduced duration was recorded in the last procedure compared with the first one. The implantation process induced a deterioration in hearing thresholds, which fully recovered after activation of the device. A post-operative hearing gain could be measured in all 3 patients: in this regard, the perceived quality of sound was shown to be better than could be expected by the measurable hearing gain.

In March 2010, the FDA approved the the Esteem, an implantable hearing system, for the treatment of patients with moderate-to-severe sensorineural hearing loss. The Esteem system consists of external testing and programming instruments and 3 implantable components:

The sensor senses vibrations from the eardrum and middle ear bones and converts these mechanical vibrations into electrical signals, which are then transmit to the sound processor, which amplifies and filters the signal to compensate for the individual patient's hearing loss. The driver converts the enhanced electrical signal back to vibrations, which are then transmitted into the inner ear where they are perceived as sound. The system is designed to alleviate the effects of hearing loss in patients aged 18 years and older. Other criteria for implanting the Esteem entail a normally functioning Eustachian tube, normal middle ear anatomy, as well as stable bilateral sensorineural hearing loss. In a multi-center clinical study of Esteem versus pre-implant hearing aids, 93 % of Esteem recipients scored equal to or better than their pre-implant hearing aids on a speech intelligibility test; 7 % scored less than with their pre-implant hearing aids, and 56 % scored better than with their pre-implant hearing aids. There are some side effects associated with the implantation of the device - 7 % of participants experienced facial paralysis, and 42 % experienced taste disturbance. The majority of these side effects resolved during the 1-year study period.

Bittencourt et al (2014) noted that the complaints associated with the use of conventional amplifying hearing aids prompted research at several centers worldwide that ultimately led to the development of implantable devices for aural rehabilitation. These investigators reviewed the history, indications, and surgical aspects of the implantable middle ear hearing devices. Semi-implantable hearing aids (e.g., the Vibrant Soundbridge system and the Maxum system), as well as fully implantable hearing aids (e.g., the fourth-generation of Carina prosthesis and the Esteem device), have their own peculiarities on candidacy and surgical procedure. The authors concluded that implantable hearing aids, which are currently in the early stages of development, will unquestionably be the major drivers of advancement in otologic practice in the 21st century, improving the quality of life of an increasingly aged population, which will consequently require increased levels of hearing support.

Free-Floating Piezoelectric Microphone in an Implantable Hearing Aid

Einger and colleagues (2016) noted that current hearing aids and implants rely on feedback compensation to prevent instability (e.g., howling), usually in the form of a digital or analogue filter. These researchers investigated the effect of mechanically stabilizing a piezo-driven mechanical amplifier inserted into the incudo-stapedial joint gap. They examined if this is possible and discerned the advantages and disadvantages of the design. These investigators examined a 10:1 scale model of a prospective implantable hearing aid comprising 1 piezoelectric sensor and 2 independent piezoelectric actuators in a single-titanium housing. As expected, the maximum gain of the device was limited by feedback between sensor input and the output of the primary actuator. The secondary actuator was used to provide a counter force to the recoil of the primary output piezo.

This added a virtual mass to the device, effectively reducing feedback in the mechano-acoustic path. The compensation unit (CU) described in this study was driven by a real-time adaptive control algorithm. Using the above approach, these researchers observed an added stable gain of greater than 30 dB, and reported a functional hearing gain of up to 40 dB. Physical and digital feedback compensation can be employed in parallel for best

results. The experimental data were backed by computer simulations. The authors concluded that these findings compared favorably with previous studies of a 2-piezo transducer with digital feedback control and showed the potential for the transducer as a hearing aid for high-frequency hearing loss.

Bilateral Versus Unilateral Hearing Aids for Bilateral Hearing Impairment in Adults

In a Cochrane review, Schilder and colleagues (2017) evaluated the effects of bilateral versus unilateral hearing aids in adults with a bilateral hearing impairment. The Cochrane ENT Information Specialist searched the ENT Trials Register; Cochrane Register of Studies Online; PubMed; Ovid Embase; CINAHL; Web of Science; ClinicalTrials.gov; ICTRP and additional sources for published and unpublished trials. The date of the search was June 8, 2017. Randomized controlled trials (RCTs) comparing the fitting of 2 versus 1 ear-level acoustic hearing aids in adults (over 18 years) with a bilateral hearing impairment, both ears being eligible for hearing aids were selected for analysis. The investigators used the standard methodological procedures expected by Cochrane. The primary outcomes were patient preference for bilateral or unilateral aids, hearing-specific health-related quality of life (QOL) and adverse effects (pain or discomfort in the ear, initiation or exacerbation of middle or outer ear infection). Secondary outcomes included: usage of hearing aids (as measured by, for example, data logging or battery consumption), generic health-related QOL, listening ability and audiometric benefit measured as binaural loudness summation. These researchers used GRADE to assess the quality of the evidence for each outcome. These researchers included 4 cross-over RCTs with a total of 209 participants, ranging in age from 23 to 85 and with a preponderance of men. All the studies allowed the use of hearing aids for a total period of at least 8 weeks before questions on preference were asked. All studies recruited patients with bilateral hearing loss but there was considerable variation in the types and degree of SNHL that the participants were experiencing; 3 of the studies were published before the mid-1990s whereas the 4th study was published in 2011. Therefore, only the most recent study used hearing aids incorporating technology comparable to that currently readily available in high-income settings. Of the 4 studies, 2 were conducted in the UK in National Health Service (NHS - public sector) patients: 1 recruited patients from primary care with hearing loss detected by a screening program whereas the other recruited patients who had been referred by their primary care practitioner to an otolaryngology department for hearing aids. The other 2 studies were conducted in the US: 1 study recruited only military personnel or veterans with noise-induced hearing loss whereas about 50 % of the participants in the other study were veterans. Only 1 primary outcome (patient preference) was reported in all studies. The percentage of patients who preferred bilateral hearing aids varied between studies: this was 54 % (51 out of 94 participants), 39 % (22 out of 56), 55 % (16 out of 29) and 77 % (23 out of 30), respectively. These investigators did not combine the data from these 4 studies. The evidence for this outcome was of very low quality. The other outcomes of interest were not reported in the included studies. The authors concluded that this review identified only 4 studies comparing the use of 1 hearing aid with 2. The studies were small and included participants of widely varying ages. There was also considerable variation in the types and degree of SNHL that the participants were experiencing. For the most part, the types of hearing aid evaluated would now be regarded, in high-income settings, as "old technology", with only 1 study looking at "modern" digital aids. However, the relevance of this was uncertain, as this review did not evaluate the differences in outcomes between the different types of technology. These investigators were unable to pool data from the 4 studies and the very low quality of the evidence led them to conclude that they did not know if people with hearing loss have a preference for 1 hearing aid or 2. Similarly, these researchers did not know if hearing-specific health-related QOL, or any of other outcomes, were better with bilateral or unilateral hearing aids.

Early Vocabulary Development Between Children with Hearing Aids and Children with Cochlear Implant

Percy-Smith and colleagues (2018) evaluated the implementation of a Nordic auditory verbal (AV) intervention for children with all degrees and types of hearing impairment (HI) using all kinds of hearing technology. A 1st specific objective was to identify differences and similarities in early vocabulary development between children with cochlear implant (CI) compared with children with hearing aids (HAs)/bone anchored hearing aids (BAHA) enrolled in a 3-year AV program, and to compare the group of children with HI to a control group of children with normal hearing (NH). A 2nd specific

objective was to study universal neonatal hearing screening (UNHS) using the 1-3-6 Early Hearing Detection and Intervention (EHDI) guidelines. A longitudinal and comparative study design was used involving 2 cohorts of children (36 children with CI and 19 children with HA/BAHA). The children were the first in Denmark to receive a 3-year AV intervention by formally trained AV-practitioners. Children were tested annually with standardized speech and language tests (i.e., Peabody Picture Vocabulary test, Reynell test and a Danish test for active vocabulary, Viborgmaterialet). Categorical variables were compared using Fischer's exact test and continuous variables were compared using Wilcoxon-Mann-Whitney test, as data were not normally distributed. Median age of diagnosis was 6 months and median age at intervention was 13 and 12 months, respectively.

There was no statistically significant difference between the 2 groups in terms of scores according to age equivalency for the 3 tests. However, there was a significant difference between children with HI regardless of hearing technology and children with NH. The authors concluded that children with HI progressed over a 3-year period, but they did not reach the same level as children with NH. The high completion rate of 98,2 % of families over a 3-year period indicated the relevance of AV practice in a Nordic country. They noted that children were diagnosed later than 3 months and intervention also started later than recommended; findings that warrant further investigation.

Treatment of Tinnitus

Sereda and colleagues (2018) noted that tinnitus affects 10 % to 15 % of the adult population, with about 20 % of these experiencing symptoms that negatively affect QOL.

In England alone there are an estimated 3/4 million general practice consultations every year where the primary complaint is tinnitus, equating to a major burden on healthcare services. Clinical management strategies include education and advice, relaxation therapy, tinnitus retraining therapy (TRT), cognitive behavioral therapy (CBT), sound enrichment using ear-level sound generators or HAs, and drug therapies to manage co-morbid symptoms such as insomnia, anxiety or depression. Hearing aids, sound generators and combination devices (amplification and sound generation within one device) are a component of many tinnitus management programs and together with information and advice are a first-line management in audiology departments for someone who has tinnitus. In a Cochrane review, these investigators examined the effects of sound therapy (using amplification devices and/or sound generators) for tinnitus in adults. The Cochrane ENT Information Specialist searched the Cochrane ENT Register; Central Register of Controlled Trials (CENTRAL, via the Cochrane Register of Studies); Ovid Medline; Ovid Embase; CINAHL; Web of Science; ClinicalTrials.gov; ICTRP and additional sources for published and unpublished trials. The date of the search was July 23, 2018; RCTs recruiting adults with acute or chronic subjective idiopathic tinnitus were selected for analysis.

These researchers included studies where the intervention involved HAs, sound generators or combination HAs and compared them to waiting list control, placebo or education/information only with no device. They also included studies comparing HAs to sound generators, combination HAs to HAs, and combination HAs to sound generators. These investigators used the standard methodological procedures expected by Cochrane. The primary outcomes were tinnitus symptom severity as measured as a global score on multi-item tinnitus questionnaire and significant adverse effects as indicated by an increase in self-reported tinnitus loudness. The secondary outcomes were depressive symptoms, symptoms of generalized anxiety, health-related QOL and adverse effects associated with wearing the device such as pain, discomfort, tenderness or skin irritation, or ear infections. They used GRADE to assess the quality of evidence for each outcome.

This review included 8 studies (with a total of 590 participants); 7 studies examined the effects of HAs, 4 combination HAs and 3 sound generators; 7 studies were parallel-group RCTs and 1e had a cross-over design. In general, risk of bias was unclear due to lack of detail about sequence generation and allocation concealment. There was also little or no use of blinding. No data for the outcomes were available for any of these researchers' 3 main comparisons (comparing HAs, sound generators and combination devices with a waiting list control group, placebo or education/information only). Data for the additional comparisons (comparing these devices with each other) were also few, with limited potential for data pooling. Hearing aid only versus sound generator device only – 1 study compared patients fitted with sound generators versus those fitted with HAs and found no difference between them in their effects on our primary outcome, tinnitus symptom severity measured with the Tinnitus Handicap Inventory (THI) at 3, 6 or 12 months (low-quality evidence). The use of both types of device was associated with a clinically significant reduction in tinnitus symptom severity. Combination HA versus HA only – 3 studies compared combination HAs with HAs and measured tinnitus symptom severity using

the THI or Tinnitus Functional Index. When these investigators pooled the data they found no difference between them (standardized mean difference [SMD] -0.15, 95 % confidence interval [CI]: -0.52 to 0.22; 3 studies; 114 participants) (low-quality evidence). The use of both types of device was again associated with a clinically significant reduction in tinnitus symptom severity. Adverse effects were not assessed in any of the included studies. None of the studies measured the secondary outcomes of depressive symptoms nor depression, anxiety symptoms or generalized anxiety, or health-related QOL as measured by a validated instrument, nor the newly developed core outcomes tinnitus intrusiveness, ability to ignore, concentration, quality of sleep and sense of control. The authors concluded that there was no evidence to support the superiority of sound therapy for tinnitus over waiting list control, placebo or education/information with no device. There was insufficient evidence to support the superiority or inferiority of any of the sound therapy options (HA, sound generator or combination HA) over each other. The quality of evidence for the reported outcomes, assessed using GRADE, was low. Using a combination device, HA or sound generator might result in little or no difference in tinnitus symptom severity. These researchers stated that future research into the effectiveness of sound therapy in patients with tinnitus should use rigorous methodology. Randomization and blinding should be of the highest quality, given the subjective nature of tinnitus and the strong likelihood of a placebo response. The CONSORT statement should be used in the design and reporting of future studies. These investigators also recommended the use of validated, patient-centered outcome measures for research in the field of tinnitus.

Hearing Rehabilitation After Obliteration of Troublesome Mastoid Cavities

In a retrospective, chart analysis, Geerse and colleagues (2020) examined post-operative hearing threshold following revision surgery and obliteration of troublesome canal wall down mastoidectomy cavities (CWDMCs). These investigators also evaluated the ability to use and tolerate conventional HAs (CHAs). This study included 249 patients with chronically draining CWDMCs who underwent revision surgery including obliteration of the mastoid cavity between 2007 and 2017 at the AMC location of the Amsterdam University Medical Centers (Amsterdam UMC). Patient characteristics, pre- and post-operative Merchant grade, surgical outcomes, pre- and post-operative hearing thresholds, and the ability/necessity to use a CHA or the ability/necessity to use a bone conduction device (BCD) were recorded. Dry ears were found in 95 % of the total cohort. Residual disease was detected in 1.6 % during MRI follow-up with no residual cholesteatoma in the obliterated area. In 3.2 % of the patients, recurrent disease was found. A significant improvement in mean air conduction level, mean bone conduction level, and mean air-bone gap (ABG) was found post-operatively ($p < 0.05$). For all types of ossicular chain reconstruction, a significant improvement in mean pure tone average was observed ($p < 0.05$). The percentage of patients with an indication for CHA was similar pre- and post-operatively (67 % both pre- and post-operatively). The ability to use a CHA improved from 3 % pre-operatively to 57 % post-operatively ($p < 0.001$). The authors concluded that the findings of this study showed that revision surgery and obliteration of CWDMCs enable successful CHA rehabilitation post-operatively. Upon this type of surgery, hearing thresholds improved significantly, but the need for rehabilitation with a CHA remains necessary in most cases.

Hearing Aids for Improvement of Depression and Cognitive Decline in the Elderly

In a retrospective, cohort study, Mahmoudi and co-workers (2019) examined the association between the use of HAs and time to diagnosis of Alzheimer disease (AD) or dementia, anxiety or depression, and injurious falls among the elderly, aged 66 years and older, within 3 years of HL diagnosis. These researchers used 2008 to 2016 national longitudinal claims data (based on office visit, inpatient, or outpatient healthcare encounters) from a large private payer. They used Kaplan-Meier curves to examine unadjusted disease-free survival (DFS) and crude and adjusted Cox regression models to examine associations between HAs and time to diagnosis of 3 age-related/HL-associated conditions within 3 years of HL diagnosis. All models were adjusted for age, sex, race/ethnicity, census divisions, and prior diagnosis of cardiovascular conditions, hypertension, hypercholesterolemia, obesity, and diabetes. Subjects included 114,862 adults, aged 66 years and older, diagnosed with HL. Measurements included diagnosis of AD or dementia; depression or anxiety; and injurious falls. Large sex and racial/ethnic differences exist in HA use. Approximately 11.3 % of women versus 13.3 % of men used HAs (95 % CI: -0.024 to -0.016). Approximately 13.6 % of Whites (95 % CI: 0.13 to 0.14) versus 9.8 % of Blacks (95 % CI: 0.09 to 0.11) and 6.5 % of Hispanics (95 % CI: 0.06 to

0.07) used HAs. The risk-adjusted hazard ratios (HRs) of being diagnosed with AD/dementia, anxiety/depression, and injurious falls within 3 years after HL diagnosis, for those who used HAs versus those who did not, were 0.82 (95 % CI: 0.76 to 0.89), 0.89 (95 % CI: 0.86 to 0.93), and 0.87 (95 % CI: 0.80 to 0.95), respectively. The authors concluded that the use of HAs was associated with delayed diagnosis of AD, dementia, depression, anxiety, and injurious falls among the elderly with HL. Although these researchers have shown an association between use of HAs and reduced risk of physical and mental decline, randomized trials are needed to determine if, and to what extent, the relationship is causal.

The authors stated that this study had several drawbacks. First, inherent limitations of using claims data include a lack of information regarding patients' socio-economic status, lifestyle choices, educational attainment, and other salient factors. While appropriate use of HAs might delay the diagnosis of age-related conditions, HA users could have different lifestyle choices and resources available to them that could also contribute to a delayed diagnosis of these conditions. Second, since these investigators used claims and diagnostic codes to define HL, they may be unable to identify all patients with HL. HL patients without official diagnoses may not be included; others may have been incorrectly included as new patients if their diagnosis preceded the 12-month look-back period. Furthermore, claims data do not include direct audiometric measurements of HL. These researchers were able to identify those who were diagnosed with HL but could not determine HL severity. They assumed that most adults, aged 66 years or older, who were diagnosed with HL had moderate-to-severe HL, and thus were in need of HAs. Third, these investigators did not have any way to measure frequency and duration of HA use, if any, among individuals who purchased HAs. Frequently, HAs were not fitted properly, and individuals did not use their HAs consistently. There are also cultural taboos among many sub-populations, such as minorities and women, regarding the use of HAs. Individuals with HL may purchase HAs on the advice of physicians or family members, but rarely use them. Fourth, analysis of falls is complex. Using claims data, these researchers controlled for no history of fall-related injuries during the 12-month period before index HL diagnosis but were not able to control for other fall-related factors. Finally, these data came from a private insurance database that might introduce biases into these findings associated with the health status and higher functioning of Medicare managed care patients. Furthermore, state-level market penetration from a single large private payer varies; thus, in the absence of sampling weights, national prevalence is not estimable at this time.

Brewster and colleagues (2020) noted that age-related HL (ARHL) is a prevalent condition associated with increased risk for depression and cognitive decline. In a 12-week, prospective, double-blind, pilot RCT, these researchers examined the feasibility of HAs for improving the mood and cognition of the elderly with ARHL. Individuals ($n = 13$) aged greater than or equal to 60 years with major depressive disorder (MDD) or persistent depressive disorder (PDD) and at least mild HL (pure tone average of greater than or equal to 30 dB) were randomized to receive full-amplification (active) versus low-amplification (sham) HAs added to psychiatric treatment as usual. Duration of HA use in hours/day, adverse events (AEs) frequency, attrition rate, and maintenance of the study blinding were the primary outcome measures. Compliance with HAs was excellent (more than 9 hours/day for both groups) and rates of AEs and drop-outs did not differ between groups. Preliminary data demonstrated differential improvement for active versus sham HAs on hearing functioning (Hearing Handicap Inventory for the Elderly [non-parametric effect size (np -ES) = 0.62]), depressive symptoms (Inventory for Depressive Symptomatology [np -ES = 0.31]), cognition (Repeatable Battery for the Assessment of Neuropsychological Status Immediate Memory [np -ES = 0.25]), and general functioning (World Health Organization Disability Assessment Schedule [np -ES = 0.53]). Significantly greater than 50 % of both groups correctly guessed their treatment assignment, indicating incomplete concealment of treatment allocation. The authors concluded that this pilot RCT for ARHL and late-life depression was feasible to execute and showed clinical promise; however, improved methods of blinding the experimental treatments are needed.

These researchers stated that larger studies should examine if hearing remediation may be an effective preventative and/or therapeutic strategy for late-life depression and cognitive decline.

The authors stated that this study had several drawbacks. First, the small sample size ($n = 13$) in this pilot study limited the authors' ability to calculate accurate effect sizes and yielded differences between the active and sham HA groups that were unreliable.

Second, the naturalistic study design limited the specific interpretations that could be made regarding the therapeutic value of HAs for depression. Almost 50 % of the subjects

in each treatment arm started a new anti-depressant treatment, and while the rates of treatment initiation were not different between groups, this may have contributed to the symptomatic and functional benefits observed. Third, the sham HA group was significantly older, which may have influenced the depressive and cognitive outcomes. Fourth, blinding of treatment assignment failed for study participants, so differential placebo effects operative between the active and sham conditions may have contributed to the results observed.

Powell and associates (2021) noted that with the increasing number of the elderly around the world, the overall number of dementia cases is expected to increase dramatically in the next 40 years. In 2020, nearly 6 million individuals in the U.S. lived with AD with anticipated growth to nearly 14 million by year 2050. This increasing prevalence, coupled with high societal burden, makes prevention and intervention of dementia a medical and public health priority. Clinicians and researchers will continue to see more individuals with HL with other co-morbidities including dementia. Epidemiologic evidence suggests an association between HL and increased risk of dementia, presenting opportunity for targeted intervention for HL to play a fundamental role in dementia prevention.

These researchers summarized current research on the association between HL and dementia and reviewed potential casual mechanisms behind the association (e.g., sensory-deprivation hypothesis, information-degradation hypothesis, common cause). They emphasized key areas of research that might best inform research of this potential casual association. These selected research priorities include examination of the causal mechanism, measurement of co-existing HL and cognitive impairment, and potential of aural rehabilitation. The authors concluded that addressing these research gaps and how results are then translated for clinical use is paramount for dementia prevention and overall health of the elderly.

Sanders et al (2021) stated that dementia currently affects 50 million people globally with this expected to triple by 2050. Even though HL is associated with cognitive decline, the underlying mechanisms are not fully understood. Considering HL is the largest modifiable risk factor for developing dementia, it is essential to study the effect of HAs on cognitive function. These investigators systematically reviewed the existing literature to examine the evidence for using HAs intervention as a treatment for deteriorating cognitive function. They carried out searches of PubMed, Cochrane Library, Embase and grey literature and located 3,060 unique records between 1990 and 2020. Two reviewers independently selected longitudinal studies observing the effects of HAs on cognitive function in individuals without dementia at onset of the study. Due to the heterogeneity of the data, a meta-analysis could not be performed. These researchers identified 17 unique studies, spanning 30 years of research and 3,526 participants. The included studies made use of 50 different cognitive function tests. These tests were grouped into separate cognitive domains according to the DSM-V classification for further analysis. The most beneficial impact of HAs appeared to be in the cognitive domain of executive function, with 6 studies showing improvement, 2 studies being inconclusive and 3 studies not demonstrating a significant effect; 3 of the 5 studies demonstrated significant improvement when screening for brief mental status. The least beneficial impact was observed in domain of complex attention, with 8 studies showing no significant effects, compared with 1 demonstrating improvement with intervention. The authors concluded that there is controversy regarding the effects of HAs on cognition. These investigators stated that additional research through randomized clinical trials with standardized cognitive assessment and longer follow-up is needed to further examine this relationship.

These researchers stated that regarding future research; first, there is a need for long-term, large-scale observational studies to better understand the effects of HAs. Cognitive decline can only be adequately observed in studies that span multiple years, if not decades. Ideally, this would be researched in the form of a randomized clinical trial. In this regard, the ongoing ACHIEVE Trial is a study of 850 elderly randomized to either a hearing intervention or successful aging intervention (an interactive health education program) to study the effects on cognition. Second, as is clear from the wide variety of cognitive measures discussed in this review, there is a need for a standardization of well-justified cognitive measures to use for this purpose. Third, further elucidation of the underlying mechanisms is necessary, and an important step for further research is the stratification of results according to subtypes of dementia. Fourth, even though domains of social and emotional cognition have not been focused on in wider literature, it would be relevant to examine the effects of HAs on these as well. Finally, as the 2020 Lancet Commission on dementia prevention, intervention, and care recognized HL to be a mid-life risk factor for dementia and considering HL starts around

mid-life, it would be interesting to conduct studies in slightly younger adults.

Brewster et al (2022) noted that recent research has revealed important neural and psychiatric consequences of HL in the elderly. In a 12-week, double-blind, randomized controlled, pilot study, these researchers examined the neural effects of HL and the impact of HAs on neuropsychiatric outcomes in older adults with major depressive disorder (MDD). A total of 25 subjects (aged 60 years or older) with MDD and moderate-profound HL were randomized to receive HAs (100 % gain targets) or sham HAs (flat 30 dB HL) in addition to psychiatric treatment-as-usual. Outcome measures included depressive symptoms (Hamilton Rating Scale for Depression [HRSD]), executive functioning (NIH Toolbox Flanker), integrity of auditory brain areas (structural MRI, diffusion tensor imaging). At baseline, worse speech discrimination was associated with auditory cortical thinning (left anterior transverse temporal gyrus: $r = 0.755$, $p = 0.012$) and lower integrity of the superior longitudinal fasciculus (FA: Left $r = 0.772$, $p = 0.025$, Right $r = 0.782$, $p = 0.022$). After 12-weeks, HAs were effective at improving hearing functioning (Hearing Handicap for the Elderly: active -12.47 versus sham -4.19, $t = -2.64$, $df = 18$, $p = 0.016$) and immediate memory (active +14.9 versus sham +5.7, $t = 2.28$, $df = 16$, $p = 0.037$). Moderate improvement was observed for HAs on executive functioning but did not reach statistical significance (Flanker: active +4.8 versus sham -2.4, $t = 1.95$, $df = 15$, $p = 0.071$). No significant effect on depression was found (HRSD: active -5.50 versus sham -7.32, $t = 0.75$, $df = 19$, $p = 0.46$). The authors concluded that HL could affect brain regions important for auditory and cognitive processing, and hearing remediation may have beneficial effects on executive functioning in MDD. Moreover, these researchers stated that further investigation with larger studies may examine if HAs may improve cognitive and depressive symptoms in older adults with MDD.

The authors stated that these findings must be considered in light of several major drawbacks, the most notable of which was the small sample size ($n = 25$), which precluded detection of small-to-medium treatment effect sizes on the order these investigators observed. Given the small sample size, the exploratory analyses did not control for multiple comparisons and larger studies are needed to replicate these findings given the possibility of type I error. Furthermore, as the study only included subjects with moderate-profound HL and MMSE above 24, results may not be generalizable to individuals with only mild HL or dementia. Finally, incomplete blinding was achieved with the use of sham HAs, and methods for improving the concealment of active versus sham HAs should be examined. While the active HA group was adequately blinded, the inadequate treatment concealment in the sham HA group may have led to negative expectations of treatment and have resulted in a placebo effect (e.g., worse hearing, cognitive, and depressive outcomes). Mitigating this drawback was the fact that the active condition was blinded in this study (i.e., subjects assigned to active HAs correctly guessed their group at roughly 50/50), minimizing a potential increase in the placebo effect component of response attributable to unblinding in this group.

Tsimpida et al (2022) stated that the adverse impact of HL extends beyond auditory impairment and may affect the individuals' psychosocial well-being. These researchers examined if there exists a causal psychosocial pathway between HL and depression in later life, via socio-economic factors and QOL, and whether HAs usage would alleviate depressive symptoms over time. They examined the longitudinal relationship between HL and depressive symptoms (CES-D) applying dynamic cross-lagged mediation path models.

These investigators used the full dataset of participants aged 50 to 89 years (74,908 person-years), from all 8 Waves of the English Longitudinal Study of Ageing (ELSA).

Their QOL (CASP-19) and their wealth were examined as the mediator and moderator of this relationship, respectively. Subgroup analyses examined differences among those with HAs within different models of subjectively and objectively identified HL. All models were adjusted for age, sex, retirement status and social engagement. Socio-economic position (SEP) influenced the strength of the relationship between HL and depression, which was stronger in the lowest versus the highest wealth quintiles. The use of HAs was beneficial for alleviating depressive symptoms. Those in the lowest wealth quintiles experienced a lower risk for depression after the use of HAs compared to those in the highest wealth quintiles. The authors concluded that HL posed a substantial risk for depressive symptoms in older adults, especially those who experienced socio-economic inequalities. These researchers stated that the early detection of HL and provision of HA may not only promote better-hearing health but could also enhance the psychosocial well-being of older adults, particularly those in a lower SEP. Moreover, these investigators stated that future studies should examine the potential effectiveness of the inclusion of screening for HL in the routine geriatric evaluation guidelines in the enhancement of psychosocial health of older adults, especially those in lower wealth

groups.

The authors stated that the findings of this study must be interpreted considering several drawbacks. First, the analyses were inevitably restricted to include only the available variables in the ELSA datasets; thus, omitted variable bias may occur, as these researchers never know whether all relevant predictors have been included in their models, and one of those left out may determine the value of an endogenous variable. Second, the ELSA study concentrated on individuals living in private households; therefore, individuals living in institutions (e.g., residential and nursing homes) were not included in the samples. Third, the CES-D scale does not measure the duration of symptoms; thus, DSM criteria for major or minor depression could not be applied to these data. Also, predicting the presence of clinically elevated depressive symptoms over time, as these researchers did in their study, referred less directly to symptom severity. Fourth, these investigators were aware that the self-reported data in the ELSA under-estimated objectively measured hearing problems. The sensitivity analysis these researchers run to examine potential differences in estimates of depression was possible only for ELSA Wave 7; this was the only ELSA Wave that included both self-reported as well as objective hearing measures for the same participants. Therefore, the error introduced by the self-reported measures was likely to reduce external validity, as individuals with depression may differentially report information on hearing status, and also their responses may be biased by cultural or population characteristics. Fifth, the ELSA questionnaire did not contain any information regarding dementia diagnosis at any ELSA Wave, which may impact on the examined associations. There are few variables that offer an indirect cognitive assessment, such as numerical ability, word-finding (verbal fluency) and prospective memory; however, these are not recorded in all Waves.

Therefore, these researchers decided to avoid the measurement bias from missing data, and they did not include any of these variables in the structural equation modelling.

The recently developed harmonized cognitive assessment protocol, part of the Healthy Cognitive Aging Project (HCAP) aimed to fill the afore-mentioned gap in the ELSA dataset, and discriminate between normal cognitive performance, cognitive impairment, and dementia status of the participants. Information from the ELSA-HCAP study is available from the ELSA Wave 9 and onwards and should be used in future studies to quantify the consequences of severe cognitive impairment for mental well-being.

Hearing Aids on Static and Dynamic Balance in Adults with Hearing Impairment

In a systematic review, Mahafza et al (2022) examined the peer-reviewed evidence to ascertain the effect of HAs on static and dynamic balance in adults with HI. These investigators carried out a search of the English language literature in 7 academic databases; and identified 909 relevant articles published before July 2021; 10 articles contained studies that met the inclusion criteria for this review. A total of 7 studies had measured static balance with 5 reporting improvements and 1 reporting no changes in balance with HA use. Two studies had measured dynamic balance with both reporting no changes with HA use. One study had measured both dynamic and static balance and reported no changes with HA use. The authors concluded that for adults with HI, the evidence was equivocal that amplification from HAs improved balance. These researchers stated that high quality studies examining the effect of HAs on balance in adults with HI are needed given this field is likely to develop in response to the growing population of adults with hearing and balance impairment worldwide. They stated that future research into the potential effects of HAs on balance in adults with HI should focus on adequately powered, longitudinal designs that use ecologically valid methods to better examine the effects of HAs on both static and dynamic balance in adults with hearing and balance impairment.

The authors stated that this systematic review was limited to studies published in peer-reviewed, English-language in journals only. The application of the search strategy and inclusion criteria identified only 10 studies for inclusion, although the authors were confident that these 10 studies represented all the current research addressing the question asked in the current review.

Campos et al (2023) noted that falls and their sequelae cost more than \$50 billion annually. The elderly with hearing loss are at 2.4 times greater risk of falls than their normal hearing peers. Current research is inconclusive regarding if HAs can offset this increased fall risk, and no previous studies considered if outcomes differed based on the consistency of HA use. In a cross-sectional study, individuals 60 years of age and older with bilateral hearing loss completed a survey consisting of the Fall Risk Questionnaire (FRQ) and questions about hearing loss history, HA use, and other common fall risk factors. Fall prevalence, as well as fall risk (based on FRQ score), was

compared between HA users and non-users. A separate group of consistent HA users (at least 4 hours daily use for more than 1 year) was also compared with inconsistent/non-users. Responses from 299 surveys were analyzed. Bi-variate analysis found 50 % reduced odds of experiencing a fall for HA users compared with non-users (OR = 0.50; 95 % CI: 0.29 to 0.85, $p = 0.01$). After adjusting for age, sex, hearing loss severity, and medication usage, those who reported any HA use still had lower odds of falls (OR = 0.48; 95 % CI: 0.26 to 0.90, $p = 0.02$) and lower odds of being at risk for falls (OR = 0.36; 95 % CI: 0.19 to 0.66, $p < 0.001$) than non-users. Results for consistent HA users showed an even stronger association of lowered odds of falling (OR = 0.35; 95 % CI: 0.19 to 0.67, $p < 0.001$) and lower odds of being at risk for falls (OR = 0.32; 95 % CI: 0.12 to 0.59, $p < 0.001$), suggesting a potential dose-response relationship. The authors concluded that the findings of this study suggested that use of HAs, especially consistent HA use was associated with lower odds of experiencing a fall or being classified as at risk for falls in the elderly with hearing loss.

Lavie et al (2024) stated that balance and postural control are related to hearing and hearing loss; however, whether they can be improved with HA use in the elderly is unclear. In a systematic review, these investigators examined controlled studies in which balance and hearing were tested in experienced older HA users to determine the potential effects of HA use on balance. The review carried out in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. The question, inclusion, and exclusion criteria were defined using the Population, Intervention, Control, Outcomes and Study design (PICOS) frame-work. Older adults with hearing loss and no experience with HAs, or balance tests carried out without HAs in HA users served as controls. A total of 803 studies were screened, 8 met the inclusion and exclusion criteria and were included in the final review; 5 of the 8 studies found a significant correlation between the use of HAs and the outcomes of the balance tests. The quality of the studies was limited or moderate. The authors concluded that the role of HAs in balance and postural control is unclear because of the quality of the studies and the sparse reporting of hearing status and HAs quality of fitting and use.

Behtani et al (2024) noted that recent studies suggested that sound amplification via HAs can improve postural control in adults with hearing impairments. Unfortunately, only a few studies used well-defined posturography measures to evaluate balance in adults with hearing loss with and without their HAs. Of these, only 2 examined postural control specifically in the elderly with hearing loss. These researchers examined the impact of HA use on postural control during various sensory perturbations in the elderly with age-related hearing loss. A total of 30 individuals with age-related hearing impairments and using HAs bilaterally were tested. Subjects were asked to carry out a modified clinical sensory integration in balance test on a force platform with and without HAs. The experiment was carried out in the presence of a broad-band noise ranging from 0.1 to 4 kHz presented via a loud-speaker. As expected, HA use had a beneficial impact on postural control, but only when visual and somatosensory inputs were both reduced. Data also suggested that HA use reduced the dependence on somatosensory input for maintaining postural control. The authors concluded that this finding can be of particular importance in the elderly considering the reduction of tactile and proprioceptive sensitivity and acuity often associated with aging. These results provided an additional argument for encouraging early HA fitting for individuals with hearing loss. Moreover, these investigators stated that hearing amplification is already encouraged to improve communication in ARHL, and these findings suggested that HA amplification could also be recommended to reduce the risk of falls.

The authors stated that one drawback of this trial was that only 5 subjects were confirmed to have normal peripheral vestibular function, highlighting the significant influence of age on various vestibular measures for adults aged over 40 years. Research specifically aimed at evaluating postural control for individuals with normal vestibular function and for those with age-related vestibular loss is needed. In the present study, incomplete, noisy or not replicable results were treated as “fail”. Since vestibular status may influence the results, it would be informative to obtain a clear diagnosis of vestibular function. Furthermore, there is a need for future studies to examine if a significant improvement of postural control when using HAs occurs for elderly individuals without vestibular dysfunction. The enhancement of postural control may be more pronounced for adults with hearing loss who also experience deterioration in vestibular function relative to those with normal vestibular function

Comparison of Transcutaneous and Percutaneous Implantable Hearing Devices

In a systematic review and meta-analysis, Gutierrez et al (2024a) compared QOL outcomes of percutaneous and transcutaneous bone conduction devices (pBCD and tBCD, respectively). Databases reviewed included PubMed, Scopus, and CINAHL. These investigators carried out a systematic review searching for English language articles from inception to March 15, 2023. Studies reporting QOL outcomes measured using a validated tool following implantation of either pBCDs or tBCDs were considered for inclusion. QOL outcomes included scores for Glasgow Benefit Inventory, Glasgow Children's Benefit Inventory, Abbreviated Profile of Hearing Aid Benefit, and the Speech, Spatial, and Qualities of Hearing Scale. A meta-analysis of continuous measures was performed. A total of 52 studies with 1,469 patients were included; 689 patients were implanted with pBCDs, and the remaining 780 were implanted with tBCDs. Average Glasgow Benefit Inventory scores for the tBCD group (33.0, 95 % CI: 22.7 to 43.3) were significantly higher than the pBCD group (30.9, 95 % CI: 25.2 to 36.6] ($\Delta 2.1$ [1.4 to 2.8], $p < 0.0001$). Mean Glasgow Children's Benefit Inventory scores ($\Delta 3.9$ [2.0 to 5.8], $p = 0.0001$) and mean gain in Abbreviated Profile of Hearing Aid Benefit scores ($\Delta 5.6$ [4.8 to 6.4], $p < 0.0001$) were significantly higher among patients implanted with tBCDs than those implanted with pBCDs. Patients implanted with tBCDs also had significantly higher gains on the Speech ($\Delta 1.1$ [0.9 to 1.3], $p < 0.0001$), Spatial ($\Delta 0.8$ [0.7 to 0.9], $p < 0.0001$), and Qualities of Hearing ($\Delta 1.2$ [1.1 to 1.3], $p < 0.0001$) portions of the Speech, Spatial, and Qualities of Hearing Scale than those implanted with pBCDs. The authors concluded that patients implanted with transcutaneous devices had better QOL outcomes than those implanted with percutaneous devices.

In a systematic review and meta-analysis, Gutierrez et al (2024b) compared audiometric outcomes, complications, and revisions required for pBCD versus tBCD implantable bone-conduction devices for the treatment of hearing loss associated with congenital aural atresia (CAA). Databases reviewed included PubMed, Scopus, and CINAHL. These investigators carried out a systematic review searching for English language articles from inception to December 14, 2022. Studies reporting audiometric outcomes or complications for either pBCDs or tBCDs for the treatment of CAA were selected for inclusion. A meta-analysis of single means and meta-analysis of proportions with comparison (Δ) of weighted proportions was conducted. A total of 56 studies with 756 patients were selected for inclusion; 190 patients were implanted with pBCDs, whereas the remaining 566 were implanted with tBCDs. Mean pure-tone audiometry improvement in the pBCD group (39.1 ± 1.1 dB) was significantly higher than in the tBCD group (34.6 ± 1.6 dB; $\Delta 4.5$ dB; 95 % CI: 4.2 to 4.7 dB; $p < 0.0001$). The average improvement in speech reception threshold was 38.6 ± 2.5 dB in the percutaneous group as compared with 32.7 ± 1.6 dB in the transcutaneous group ($\Delta 5.9$ dB, 95 % CI: 5.3 to 6.5 dB, $p < 0.0001$). Overall complication rates for patients implanted with pBCDs and tBCDs were 29.0 % (15.7 % to 44.4 %) and 9.4 % (6.5 % to 13.0 %), respectively ($\Delta 19.6$ %, 95 % CI: 12.0 % to 27.7 %, $p < 0.0001$). The authors concluded that patients with CAA implanted with pBCDs exhibited significantly better audiometric outcomes than those implanted with tBCDs; however, complication rates were significantly higher among the pBCD group.

Cartilage Conduction Hearing Aids

Nishimura et al (2023) noted that a relatively loud sound is audible when a vibrator is attached to the aural cartilage; this form of conduction is known as cartilage conduction (CC). In Japan, a new type of HA has been developed using CC and has been available in clinical practice since 2017. A clinical study carried out before its launch showed its benefits, especially in patients with aural atresia who were unable to use air conduction HAs. Several studies have been published on the benefits of CC-hearing aids (CC-HAs) since their introduction into clinical practice. Most of the patients included in these studies had canal stenosis or aural atresia, and the purchase rates of CC-HAs in these patients were relatively high. However, the number of patients with canal-open ears was small, with overall poor results in the trials, with the exception of patients with continuous otorrhea. The authors concluded that CC-HAs are considered a good option for compensating for hearing loss in ears with canal stenosis or atresia in both bilateral and unilateral cases; however, CC-HAs are not currently considered the 1st-choice for patients with a canal-open ear; however, they could provide significant benefits in specific cases, such as continuous otorrhea. Moreover, these researchers stated that further studies are needed to clarify the indications for use in canal-open ears.

In a systematic review, Li et al (2023) examined the effectiveness and benefit of CC-HAs and that factors that influence purchasing decisions. The hearing thresholds and functional gain following CC-HA wear were synthesized. A 1-way analysis of variance compared the differences in the hearing thresholds and functional gain at individual

frequencies and in patients with a variety of pathological changes. The synchronized aided hearing threshold and functional gain at 2.0 kHz were significantly better than at 0.5, 1.0, and 4.0 kHz. There was no significant difference in the synchronized unaided hearing thresholds across individual frequencies between 0.5 and 4.0 kHz. The synchronized functional gain in patients with atresia was significantly greater than in patients with aural atresia or stenosis and middle ear pathologies with normal ear canals. The acceptability of CC-HAs in terms of purchase decision was influenced by the condition of the external auditory meatus and severity of hearing loss, with the highest purchase rate seen in patients with aural atresia or stenosis. The authors concluded that the findings of this systematic review revealed that CC-HAs appeared to be effective hearing devices to improve hearing thresholds, speech recognition, as well as overall hearing ability in patients with various types of hearing loss. Both the synchronized aided hearing thresholds and functional gain at 2.0 kHz were significantly better than those at the frequencies of 0.5, 1.0, and 4.0 kHz. In addition, the synchronized functional gain in patients with atresia was significantly greater than in patients with stenosis and those with middle ear pathologies but with open ear canals suggesting that CC-HAs may provide greater benefits for individuals with atresia. The acceptability in terms of purchasing the device was affected by the condition of the external auditory meatus and the severity of hearing loss. Patients with aural atresia or stenosis showed the highest purchasing rate. Although CC-HAs do not demonstrate advantages in terms of functional gain in comparison to existing bone conduction hearing devices (BCHDs), due to its fitting procedure advantages and cosmetic considerations, CC-HAs are shown as a viable and advantageous choice for some patients with aural atresia or microtia. Moreover, these investigators stated that further research is needed to evaluate the subjective benefits and satisfaction of CC-HAs using objective outcome measures. This will provide insight into patients' experiences, perceived benefits, and overall satisfaction as well as the psychological impact and emotional well-being of CC-HA users. Furthermore, a finite element analysis and the inclusion of effective gain measurements in future studies would provide valuable insights into the role of cartilage conduction for sound transmission in auditory function; thereby, further facilitating the use of CC-HAs.

Management of Chemotherapy-Induced Ototoxicity in Children

Oh and Park (2025) stated that chemotherapy-induced ototoxicity is a significant concern in pediatric cancer patients, especially those treated with platinum-based agents, such as cisplatin and carboplatin. These investigators examined its prevalence, risk factors, early diagnosis, as well as management strategies. They carried out a literature review to evaluate the effects of ototoxic chemotherapy, screening methods, as well as treatment approaches. Various grading scales and rehabilitation strategies were examined. Platinum-based chemotherapy results in ototoxic hearing loss in about 100 % of cases, including high-frequency (HF) and delayed-onset losses. Younger age, higher cumulative dose, and cranial irradiation increased the risk. Screening adherence remains sub-optimal, despite guidelines recommending early detection via HF audiometry. Sodium thiosulfate may reduce ototoxicity in non-metastatic cases. If appropriate, HAs and CIs can support communication and language development. The authors concluded that ototoxic hearing loss is a prevalent, yet under-diagnosed, complication of pediatric cancer treatment. Standardized screening, oto-protective strategies, as well as early rehabilitation are crucial to minimize their impact on language and QOL. These investigators stated that greater awareness and national guidelines are needed to improve the care of pediatric cancer survivors.

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